Abhijit Karale

RTL Design & Verification Engineer | SystemVerilog | UVM | Formal Verification
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PROFESSIONAL SUMMARY

RTL Design and Verification Engineer with strong expertise in SystemVerilog, UVM, and Formal Verification (JasperGold). Hands-on experience in 5-stage RISC-V core design, APB bus slaves, CDC-safe dual-clock FIFOs, and Cadence Xcelium. Unique dual advantage: 2+ years of embedded firmware and board bring-up allowing correlation of silicon issues to real boards.

TECHNICAL SKILLS

- HDLs & Languages: SystemVerilog, Verilog, VHDL, C/C++, Embedded C, Python, Shell Scripting
- Verification Methodologies: UVM, Constrained-Random, SVA, Functional/Code Coverage, Formal Verification (JasperGold)
- EDA Tools: Cadence Xcelium, JasperGold, Synopsys Design Compiler, QuestaSim, Vivado, ModelSim, Verdi
- Protocols & Architecture: RISC-V (RV32I), AMBA-APB, AXI4-Lite, Gray-code CDC FIFOs, UART, SPI, I2C, Low Power (UPF)

EXPERIENCE & TRAINING

The Silicon Sandbox â **Trainee - Advanced RTL Design & Verification Intern** Jan 2026 - Present | Bangalore

- Advanced industrial verification in SystemVerilog and Universal Verification Methodology (UVM).
- Developed constrained-random tests, drivers, scoreboards, and functional coverage closure in Cadence Xcelium.
- Applied Low Power Design concepts with Unified Power Format (UPF) on digital architectures.

Velaxtra Pvt Ltd â **Sr Embedded Firmware Engineer** Oct 2025 - Jun 2026 | Bangalore

- Developed embedded C firmware, peripheral drivers (UART, SPI, I2C, ADC, PWM) and FreeRTOS task scheduling.
- Led board-level defect isolation, correlating hardware timing anomalies with firmware behavior.

Minilec India Pvt Ltd â **Embedded System Engineer** Jul 2024 - Sep 2025 | Pune

- Designed and debugged STM32 / ARM Cortex-M control systems, multi-layer PCB signal integrity, and analog circuits.

Saksham Semiconductors â **Analog Circuit Design Intern** May 2024 - Jul 2024 | Hyderabad

- Characterized MOSFET behavior, designed current mirrors and differential amplifiers using SPICE simulations.

KEY RTL & VERIFICATION PROJECTS

32-bit RISC-V (RV32I) Pipelined Core & Layered UVM Environment SystemVerilog, UVM, Xcelium

- Designed 5-stage pipeline with hazard detection and operand forwarding; built layered UVM testbench achieving 95%+ coverage.

Memory Controller with Formal Property Verification (JasperGold) SystemVerilog, SVA, JasperGold

- Wrote comprehensive SVA properties for address boundaries and no-read-before-write; achieved 100% formal mathematical proofs.

Parameterizable Asynchronous FIFO with Gray-code CDC Verilog, CDC, SVA

- Implemented 2-flip-flop synchronizers, Gray pointer conversion, full/empty flag generation with zero metastability.

EDUCATION

MAEER's MIT College of Railway Engineering & Research, Barshi 2023 - 2026 | CGPA: 7.86
MAEER's MIT College of Railway Engineering & Research â **Diploma in Railway Engineering** 2021 - 2023 | 91.10%